

ClinicQ

Software Design Specification (SDS)

Project Title: ClinicQ — Clinic Appointment and Live Queue System
Course: CSE 451 — Software Engineering
Institution: Independent University, Bangladesh (IUB)
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1. Introduction

1.1 Purpose

The purpose of this Software Design Specification (SDS) is to describe the structural and behavioral design of ClinicQ, a web-based Clinic Appointment and Live Queue System. This document explains how the system's major functions, user interactions, appointment processes, queue operations, and information flows are represented through UML diagrams and a Data Flow Diagram (DFD).

The SDS is intended to provide a common understanding of the proposed system design for developers, testers, project members, instructors, and other stakeholders. It serves as a design reference during implementation, testing, and future maintenance.

1.2 Scope

ClinicQ is designed to support appointment scheduling and patient queue management for a clinic with multiple doctors. Patients can search for doctors, book appointments, manage their bookings, check in, and monitor their queue status. Receptionists can handle phone bookings, walk-in patients, queue adjustments, and patient registration-related tasks. Doctors can view their queues, call patients, conduct consultations, and record visit information. Clinic administrators can manage staff, doctors, clinic rules, and reports.

The system also supports notifications and a waiting-room display to help communicate appointment and queue information.

1.3 Intended Audience

This document is intended for:

- **Developers:** To understand the system's intended behavior and interactions.
- **Testers:** To identify important workflows, state transitions, and expected outcomes.
- **Project members:** To maintain consistency between requirements and design.
- **Course instructor and examiners:** To review the design decisions and diagram completeness.
- **Clinic stakeholders:** To understand how the proposed system supports clinic operations.

1.4 References

The primary reference for this document is the *ClinicQ Software Requirements Specification (SRS)*. The use case diagram, state diagrams, and DFD presented in this document should be consistent with the requirements and business rules specified in that SRS.

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2. System Overview

2.1 System Description

ClinicQ is a stand-alone web application intended to improve appointment scheduling and daily patient queue management. It provides a centralized system for patients and clinic staff to access relevant appointment, queue, and visit information.

The system is designed to reduce manual scheduling effort, prevent double bookings, improve visibility into waiting times, and help clinic staff manage patient flow more efficiently.

2.2 Major System Functions

The major functions of ClinicQ include:

1. Patient account registration and profile management.
2. Doctor search and appointment booking.
3. Appointment cancellation and rescheduling.
4. Appointment check-in and queue tracking.
5. Receptionist-assisted booking and walk-in registration.
6. Queue management, including patient calling, emergency handling, and removal.
7. Doctor availability and leave management.
8. Consultation management and recording of visit notes and prescriptions.
9. Patient history access, subject to authorization.
10. Staff and clinic administration.
11. Notification delivery and waiting-room queue display.
12. Reports, audit records, and patient data requests.

2.3 General System Workflow

The general workflow begins when a patient searches for a doctor and books an available appointment. On the visit day, the patient checks in and enters the appropriate queue. The receptionist can assist patients who book by phone or arrive without an online appointment.

The doctor manages the queue by calling patients and conducting consultations. After a consultation, the relevant visit information is recorded. Notifications and queue displays communicate appropriate status updates. The clinic administrator manages system settings, staff accounts, and reports.

The exact workflow depends on the user's role, appointment status, queue status, and applicable clinic rules.

3. System Actors and Responsibilities

The following actors represent the primary users and external systems interacting with ClinicQ.

Actor	Responsibilities
Patient	Registers an account, manages personal information, searches for doctors, books or manages appointments, checks in, and views queue information.
Receptionist	Handles phone bookings, walk-in patients, queue adjustments, patient information updates, and daily schedule printing.
Doctor	Views the queue, calls patients, conducts consultations, manages consultation outcomes, and records visit information.
Clinic Administrator	Manages staff accounts, doctors, clinic rules, reports, audit records, and authorized patient data requests.
Notification Service	Supports the delivery of system-generated notifications through the configured communication channels.
Waiting-Room Display	Presents relevant queue information for patients and visitors in the clinic.

The external notification service and waiting-room display are treated as external systems where appropriate. Their precise responsibilities and interfaces should follow the final use case diagram and SRS.

4. Use Case Design

4.1 Purpose of the Use Case Diagram

The Use Case Diagram represents the functional interactions between ClinicQ and its actors. It identifies the main services available to patients, receptionists, doctors, and clinic administrators, along with the external systems involved in notifications and queue display.

The diagram helps define the system boundary and clarifies which actor initiates or participates in each function. It also provides a high-level view of the functional scope before implementation.



Figure 1: Use Case Diagram of ClinicQ.

4.2 Use Case List

The following list summarizes the use cases represented in the design.

Use Case ID	Use Case Name	Primary Actor
UC-01	Register Account	Patient
UC-02	Manage Profile and Data	Patient
UC-03	Change Mobile Number	Patient
UC-04	Search Doctors	Patient
UC-05	Book Appointment	Patient
UC-06	Manage My Appointments	Patient
UC-07	Check In and View Live Queue	Patient
UC-08	View Visit History	Patient
UC-09	Book Appointment by Call	Receptionist
UC-10	Check In or Register Walk-in Patient	Receptionist
UC-11	Manage Queue	Receptionist
UC-12	Record Vitals	Receptionist
UC-13	Update Patient Mobile Number	Receptionist
UC-14	Print Daily Schedule	Receptionist
UC-15	Set Availability and Leave	Doctor
UC-16	View Queue and Call Next Patient	Doctor
UC-17	Complete or Skip Consultation	Doctor
UC-18	Record Visit Notes and Prescription	Doctor
UC-19	View Patient History	Doctor
UC-20	Manage Staff Accounts	Clinic Administrator
UC-21	Manage Doctors and Clinic Rules	Clinic Administrator
UC-22	View Reports and Audit Log	Clinic Administrator
UC-23	Process Data Requests	Clinic Administrator
UC-24	Send Notification	Notification Service

UC-25	Show Queue on Waiting-Room Display	Waiting-Room Display

4.3 Use Case Relationships

The diagram may use UML relationship types such as include and extend to represent dependencies between use cases.

- **Include:** Represents behavior that is required as part of another use case.
- **Extend:** Represents additional or conditional behavior that occurs under specified circumstances.
- **Association:** Connects an actor to a use case in which the actor participates.

These relationships should only be used when they accurately reflect the requirements. For example, a notification should be modeled as an included use case only if it is mandatory for every execution of the related use case. If notification delivery is conditional, an extension or a separate interaction may be more appropriate.

5. Use Case Narratives

This section provides brief descriptions of the main use cases. These summaries should be expanded where necessary to match the detailed narratives and confirmation criteria in the SRS.

5.1 Patient Use Cases

UC-01: Register Account
The patient provides the required registration information. The system validates the submitted information and performs the required verification process before activating the account. Invalid or incomplete information is handled according to the applicable validation rules.

UC-02: Manage Profile and Data
The patient accesses their profile and updates permitted personal information or manages available data-related options. The system validates changes and restricts access to the patient's own information.

UC-03: Change Mobile Number
The patient requests a mobile number change. The system validates the new number and performs the required verification before accepting the change.

UC-04: Search Doctors
The patient searches for doctors using available information, such as specialty or doctor name. The system presents matching results and relevant availability information where available.

UC-05: Book Appointment
The patient selects an available doctor and appointment slot and submits a booking request. The system checks slot availability and applicable booking rules before confirming the appointment. If the slot is unavailable, the patient must select another available option.

UC-06: Manage My Appointments
The patient views existing appointments and performs permitted actions, including cancellation or rescheduling. The system enforces the applicable time limits and business rules.

UC-07: Check In and View Live Queue
The patient checks in for an eligible appointment and views relevant queue information. The system updates the patient's queue status and displays available information about their position or estimated waiting time.

UC-08: View Visit History
The patient requests their previous visit information. The system displays only records the patient is authorized to access.

5.2 Receptionist Use Cases

UC-09: Book Appointment by Call
The receptionist receives an appointment request by telephone, identifies or registers the patient as required, and selects an available doctor and slot. The system validates availability and applicable rules before saving the booking.

UC-10: Check In or Register Walk-in Patient
The receptionist checks in an eligible patient or registers a walk-in patient according to clinic rules. The system records the required information and assigns the appropriate queue entry.

UC-11: Manage Queue
The receptionist monitors and manages queue entries. Depending on their permissions, the receptionist may handle emergencies, move patients, or remove entries with the required reason recorded.

UC-12: Record Vitals
The receptionist records the patient's required vital measurements when applicable. The system associates the recorded information with the correct patient and relevant visit.

UC-13: Update Patient Mobile Number
The receptionist updates a patient's mobile number when authorized. The system records the change and applies any required validation or audit rules.

UC-14: Print Daily Schedule
The receptionist requests the daily schedule for operational use. The system generates a printable view of the relevant appointment information.

5.3 Doctor Use Cases

UC-15: Set Availability and Leave
The doctor updates their availability or records leave. The system uses the updated schedule when determining appointment availability and handling affected appointments.

UC-16: View Queue and Call Next Patient
The doctor views the relevant queue and calls the next eligible patient. The system applies queue-ordering rules and prevents an additional consultation from starting when another consultation is already active, where required by the SRS.

UC-17: Complete or Skip Consultation
The doctor records the outcome of a consultation or skips a patient who does not respond. The system updates the queue state and applies the required skip-count and receptionist-review rules.

UC-18: Record Visit Notes and Prescription
The doctor records relevant consultation notes and prescription information. The system associates the information with the correct patient and visit, subject to the applicable access controls.

UC-19: View Patient History
The doctor accesses the authorized history of a patient when needed for consultation. The system restricts access according to the user's role and permissions.

5.4 Administrator Use Cases

UC-20: Manage Staff Accounts
The clinic administrator creates or manages staff accounts and their permitted access. The system applies the appropriate account and authentication rules.

UC-21: Manage Doctors and Clinic Rules
The administrator manages doctor information and configurable clinic rules. Changes affect relevant system operations according to the defined requirements.

UC-22: View Reports and Audit Log
The administrator views available operational reports and audit records. Access to these records is restricted to authorized users.

UC-23: Process Data Requests
The administrator processes authorized patient data requests according to the system's data-management and privacy requirements.

5.5 External-System Use Cases

UC-24: Send Notification
The system requests notification delivery when a supported event requires communication with a patient or other intended recipient. Delivery results or failures should be handled according to the notification integration design.

UC-25: Show Queue on Waiting-Room Display
The system provides the relevant queue information to the waiting-room display. The display presents permitted information without exposing unnecessary personal or medical details.

6. State Diagram Design

State diagrams describe the lifecycle of important entities in ClinicQ. Each diagram identifies the possible states of an entity and the events or conditions that cause transitions between those states.

6.1 Appointment State Diagram

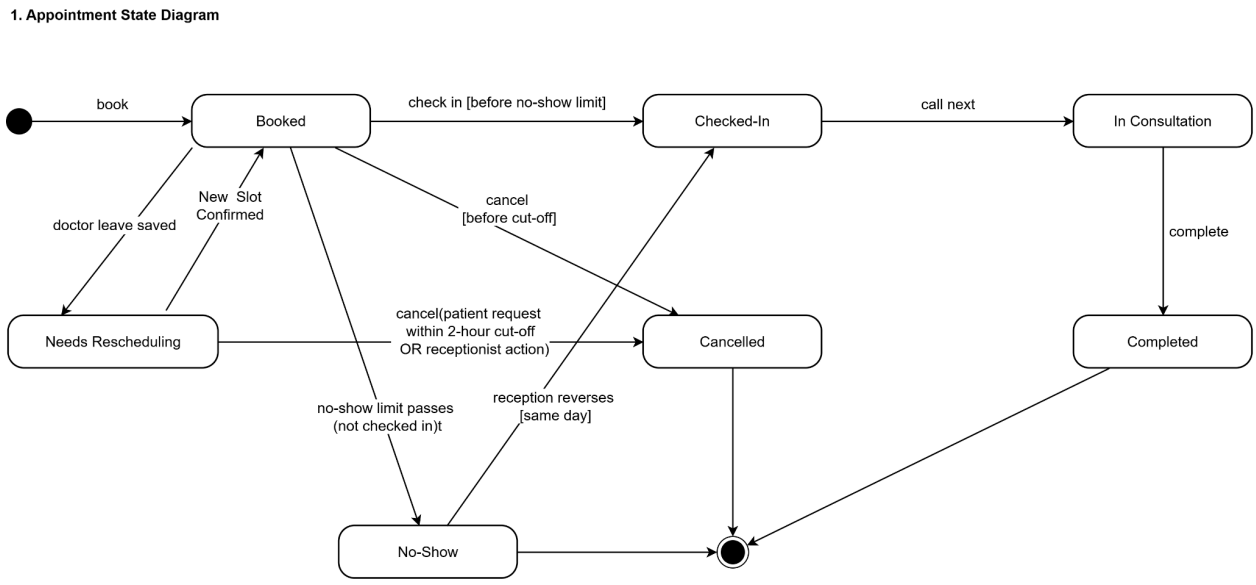


Figure 2: Appointment State Diagram.

The Appointment State Diagram represents the lifecycle of an appointment, from booking to its final outcome. It helps clarify how appointments change status as patients check in, cancel, miss appointments, or require rescheduling.

The **Booked** state represents a confirmed appointment that has not yet reached a later lifecycle state. When the patient checks in, the appointment can transition to **Checked-In**. The

appointment may subsequently be associated with an active consultation and eventually reach **Completed**.

An appointment may transition to **Cancelled** when cancellation is permitted under the applicable rules. If a doctor becomes unavailable or a scheduling conflict occurs, the appointment may enter **Needs Rescheduling** until a replacement slot is confirmed.

The **No-Show** state represents an appointment for which the patient did not check in under the defined no-show rules. Any reversal of a no-show status must follow the authorized process and applicable audit requirements.

Completed, Cancelled, and No-Show are final outcomes in this design. Needs Rescheduling is an intermediate state because a new appointment slot may still be confirmed.

6.2 Queue Entry State Diagram

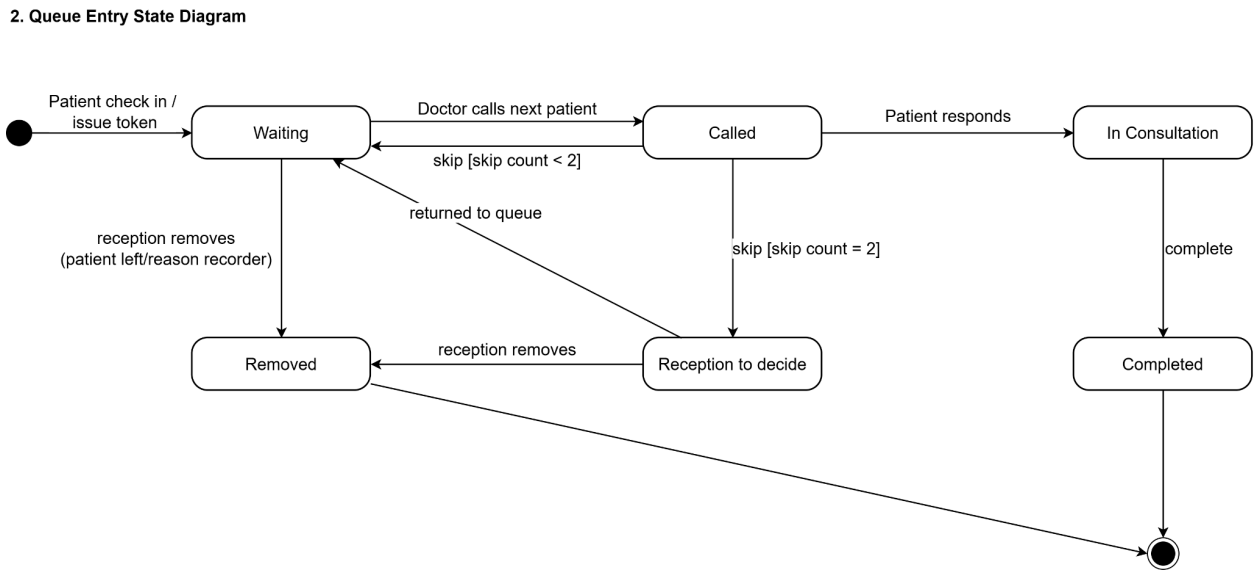


Figure 3: Queue Entry State Diagram.

The Queue Entry State Diagram describes the progress of an individual patient through the clinic's queue. It begins when a valid queue entry is created and the patient enters the **Waiting** state.

When the doctor calls the patient, the entry moves to **Called**. If the patient responds and the consultation begins, the entry moves to **In Consultation**. Once the consultation finishes, the entry moves to **Completed**.

If the patient does not respond, the system applies the configured skip-count rules. Depending on the number of unsuccessful calls, the patient may return to **Waiting** or be sent to a **Reception to Decide** state for further action.

From the decision state, reception may return the patient to the queue or remove the entry when permitted and when the required reason has been recorded. A patient may also be removed from the waiting queue through an authorized reception action.

These transitions separate appointment outcomes from queue operations. A patient leaving the queue does not automatically imply that their appointment has been cancelled.

6.3 Doctor Schedule State Diagram

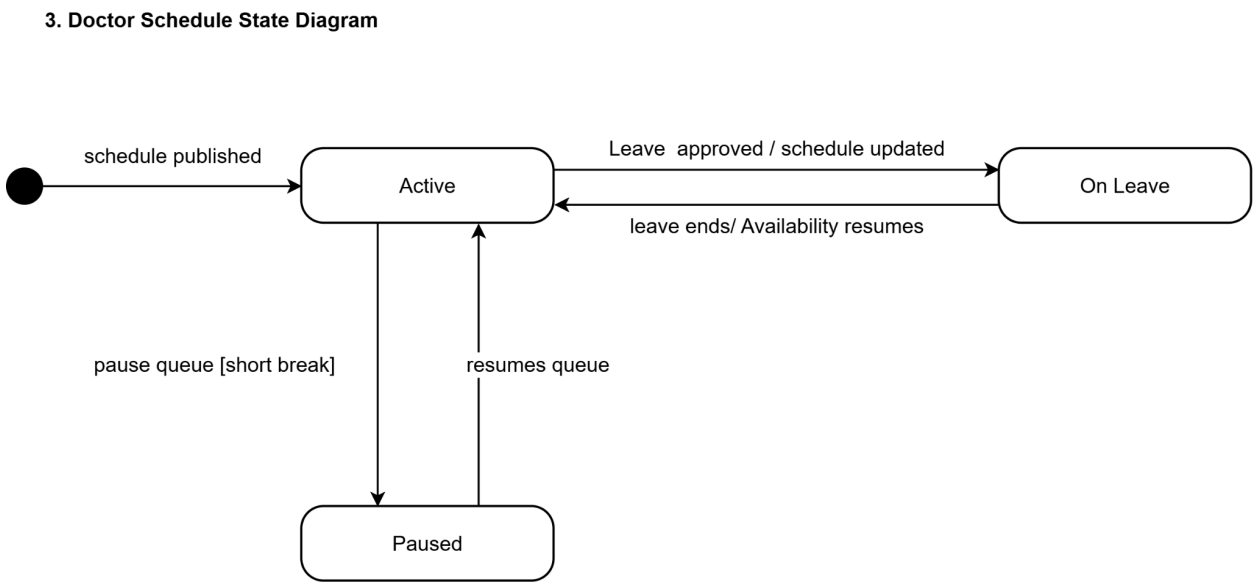


Figure 4: Doctor Schedule State Diagram.

The Doctor Schedule State Diagram represents the availability status used for appointment planning and clinic operations.

The **Active** state indicates that the doctor's schedule is available for the applicable operations. When leave is recorded or the doctor becomes unavailable, the schedule transitions to **On Leave**. When the doctor becomes available again, the schedule can return to **Active**.

These states help the system determine whether a doctor can receive new appointments and whether existing appointments may be affected by changes in availability.

Queue operation states, such as Running and Paused, should be modeled separately from doctor schedule states because temporarily pausing a queue does not necessarily mean that the doctor is on leave.

6.4 User Account State Diagram

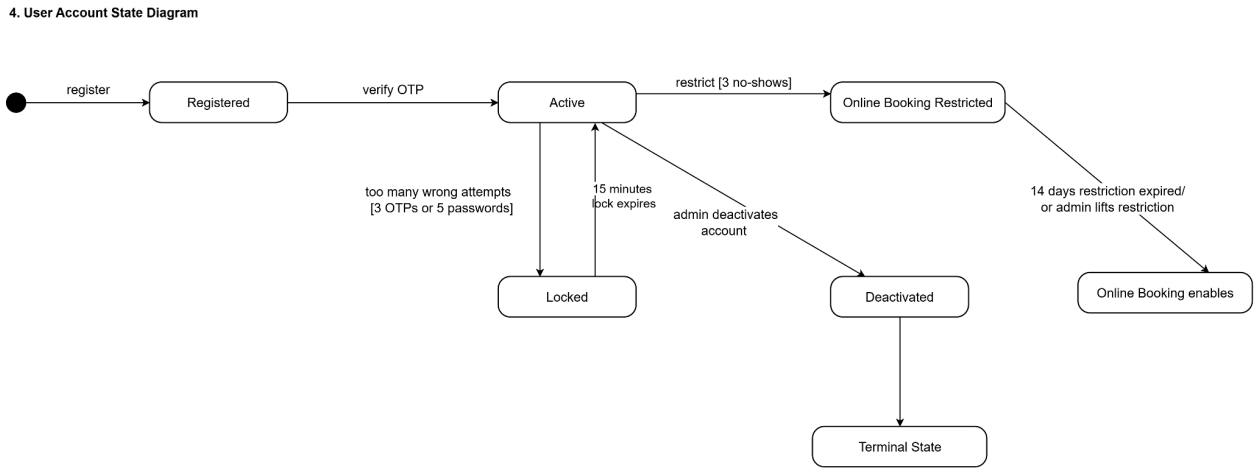


Figure 5: User Account State Diagram.

The User Account State Diagram describes important account lifecycle and access-control states.

A newly created account begins in **Registered**. Following successful completion of the required verification process, it transitions to **Active**. An account in the Active state can use the functions permitted by its role and permissions.

Repeated unsuccessful staff password attempts may cause an account to enter **Locked**. The account may return to Active when the specified temporary lock period expires, subject to the authentication rules.

The design distinguishes account status from online booking restrictions. For example, a patient's online booking access may be restricted because of repeated no-shows without necessarily deactivating the entire account. Such a restriction should be removed when the applicable restriction period expires or an authorized administrator lifts it.

Registration verification failures and staff password-lock behavior should remain separate wherever the SRS defines different attempt limits or lock periods.

7. Data Flow Diagram (DFD Level 0)

7.1 Purpose

The Data Flow Diagram represents how information moves between external entities and the ClinicQ system. It provides a high-level view of the information exchanged during appointment booking, queue management, consultation, administration, and notification delivery.

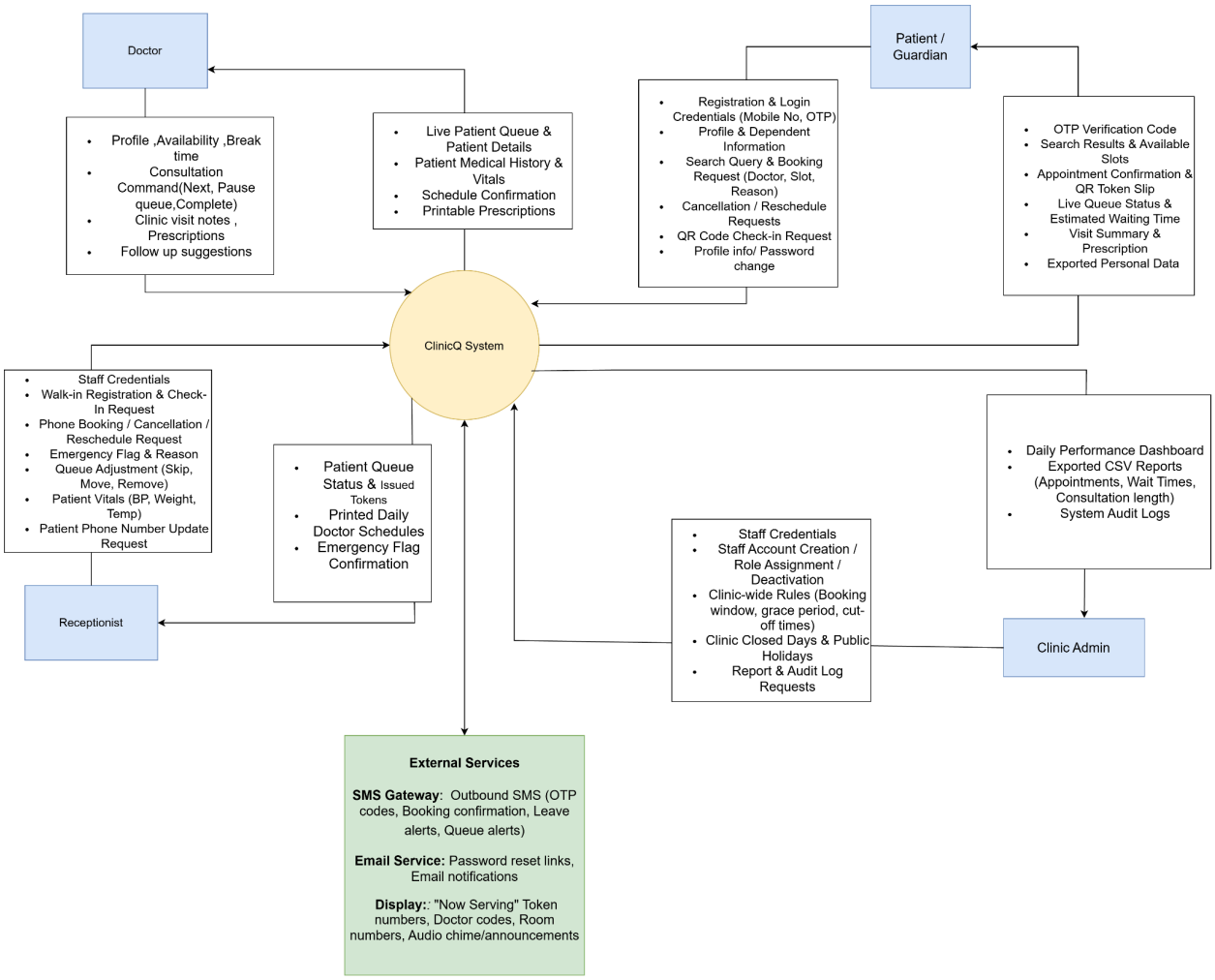


Figure 6: Data Flow Diagram Level 0 of ClinicQ.

7.2 External Entities

The main external entities represented in the diagram are:

- **Patient:** Provides registration information, search criteria, appointment requests, check-in information, and other permitted requests. Receives appointment and queue information.

- **Receptionist:** Provides assisted bookings, walk-in information, queue-management instructions, and permitted patient updates. Receives schedules and operational queue information.
- **Doctor:** Provides availability updates, queue actions, consultation outcomes, visit notes, and prescriptions. Receives appointment, queue, and authorized patient-history information.
- **Clinic Administrator:** Provides staff-management and clinic-configuration actions. Receives permitted reports and audit information.
- **Notification Service:** Receives notification requests and may return delivery status.
- **Waiting-Room Display:** Receives the permitted queue information required to display the current queue.

The exact entities should match the final DFD.

7.3 Main Data Flows

Patient-to-System Flow: Patients submit account details, doctor searches, appointment requests, cancellation or rescheduling requests, and check-in actions. ClinicQ returns relevant appointment details, confirmation information, and queue status.

Receptionist-to-System Flow: Receptionists submit assisted bookings, walk-in registrations, patient updates, and queue-management actions. ClinicQ provides daily schedules, patient information needed for authorized tasks, and current queue information.

Doctor-to-System Flow: Doctors submit availability changes, patient-calling actions, consultation outcomes, visit notes, and prescriptions. ClinicQ provides the doctor's schedule, queue information, and authorized patient history.

Administrator-to-System Flow: The administrator submits staff-management and clinic-configuration actions. ClinicQ provides permitted operational reports and audit information.

System-to-Notification-Service Flow: ClinicQ sends notification requests when applicable events occur. The external service processes the requests and may return delivery results.

System-to-Waiting-Room-Display Flow: ClinicQ supplies relevant queue information for presentation in the clinic. The information displayed must follow the system's privacy and access requirements.

7.4 DFD Interpretation

The DFD shows the principal information exchanges needed to support the ClinicQ workflow. Patient requests enter the system through the relevant interfaces, and the system processes them according to business rules and user permissions. Updated appointment and queue information is then made available to authorized users and connected external systems.

The DFD should use clearly labeled arrows to show what information is being transferred. If the course requires a strict context diagram, the entire ClinicQ system should be represented by a single process, with external entities and their data flows around it. If the submitted diagram includes internal processes or data stores, its level and notation should be labeled consistently with the instructor's requirements.

8. Data and Information Management

ClinicQ handles several categories of information to support its core functions. The following are logical information categories rather than a finalized physical database schema.

8.1 Patient Information

Patient information includes identifying and contact details required for account management, appointment scheduling, and authorized clinic operations. Access should be restricted according to the user's role and permissions.

8.2 Appointment Information

Appointment information includes the patient, doctor, appointment date and time, booking status, and other information required to manage the booking lifecycle. The system must enforce availability and booking rules to prevent conflicting appointments.

8.3 Queue Information

Queue information includes the relevant patient or appointment reference, queue order, queue status, check-in information, and required skip or removal details. It is used to determine which patient should be called and to communicate queue progress.

8.4 Doctor Schedule Information

Schedule information includes doctor availability, working times, and leave-related information. The system uses this information when presenting available appointment slots and handling schedule changes.

8.5 Visit Information

Visit information may include recorded vital measurements, consultation notes, prescriptions, and visit outcomes, according to the defined requirements. Such information must only be accessible to authorized users.

8.6 Administrative and Audit Information

Administrative information includes staff accounts, clinic rules, reports, and relevant audit records. Audit records support accountability by recording applicable actions and changes in accordance with the system requirements.

The final physical database structure, field types, primary keys, foreign keys, and relationships should be specified separately if required by the project or instructor.

9. Design Rules and Constraints

The ClinicQ design must follow the applicable business rules and non-functional requirements defined in the SRS.

9.1 Appointment Integrity

The system must validate appointment availability before confirming a booking. Concurrent booking requests must not result in multiple confirmed appointments for the same unavailable slot.

9.2 Cancellation and Rescheduling

Appointment cancellation and rescheduling must follow the permitted time limits and authorization rules. Doctor leave and schedule changes must be handled according to the defined appointment-management requirements.

9.3 Queue Integrity

Queue actions must follow the defined ordering and consultation rules. The system must track unsuccessful patient calls where required and enforce the appropriate reception-review process after the specified number of skips.

9.4 Authentication and Authorization

The system must enforce account verification, applicable login restrictions, and role-based access. Users must not be able to access or modify information beyond their permitted responsibilities.

9.5 Privacy and Auditability

Patient information, visit records, and administrative information must be protected according to the SRS. Sensitive changes and administrative actions must be recorded where the requirements call for an audit record.

9.6 Notifications and External Interfaces

Notification and display integrations must use the information required for their intended purpose. The system should handle integration failures according to the applicable requirements rather than treating every request as successfully delivered.

9.7 Consistency with Requirements

The design must remain consistent with the SRS. If a diagram suggests behavior that is not supported by the SRS, the team should verify the requirement before treating that behavior as part of the final design.

10. Requirements Traceability

Requirements traceability connects the system's functional requirements to the design elements that represent them. The following table provides a high-level mapping.

Requirement Area	Main Design Element	Expected Coverage
Account registration and access	Use Case Diagram; User Account State Diagram	Registration, verification, account status, and access restrictions
Appointment management	Use Case Diagram; Appointment State Diagram	Booking, cancellation, rescheduling, and appointment outcomes
Queue management	Use Case Diagram; Queue Entry State Diagram	Check-in, waiting, calling, consultation, skipping, and removal
Doctor availability	Use Case Diagram; Doctor Schedule State Diagram	Availability, leave, and schedule changes
Consultation management	Use Case Diagram; Queue Entry State Diagram	Consultation progress and completion
Visit information	Use Case Diagram; DFD Level 0	Authorized recording and retrieval of visit data
Notifications	Use Case Diagram; DFD Level 0	Notification requests and external delivery interaction
Waiting-room display	Use Case Diagram; DFD Level 0	Delivery of permitted queue information
Administration and reporting	Use Case Diagram; DFD Level 0	Staff management, clinic rules, reports, and audit information

Privacy and data requests	Use Case Diagram; DFD Level 0	Authorized data access and request processing

This mapping is a summary, not a substitute for requirement-level traceability. The final design review should verify the relevant SRS user story IDs and confirmation criteria against the actual diagrams and use case narratives.

11. Conclusion

This Software Design Specification describes the proposed design of ClinicQ through a Use Case Diagram, state diagrams, and a Data Flow Diagram. Together, these diagrams illustrate the main actors, functional interactions, lifecycle transitions, and information exchanges required for appointment scheduling and live queue management.

The design separates appointment status from queue status and distinguishes doctor availability from temporary queue operations. It also identifies the roles of external notification and waiting-room display systems.

The SDS provides a design reference for implementation and testing. Before implementation is considered complete, each design element must be checked against the SRS, and any missing, inconsistent, or unsupported behavior must be resolved.

12. Revision History

Version	Date	Description	Author
1.0	28 October 2026	Initial SDS draft containing the use case design, state diagrams, DFD, and design descriptions.	ClinicQ Project Team